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Low-cost evaluation of the exchange Fock matrix from Cholesky and density fitting representations of the electron repulsion integrals

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The authors propose a new algorithm, "local K" (LK), for fast evaluation of the exchange Fock matrix in case the Cholesky decomposition of the electron repulsion integrals is used. The novelty lies in the fact that rigorous upper bounds to the contribution from each occupied orbital to the exchange Fock matrix are employed. By formulating these inequalities in terms of localized orbitals, the scaling of computing the exchange Fock matrix is reduced from quartic to quadratic with only negligible prescreening overhead and strict error control. Compared to the unscreened Cholesky algorithm, the computational saving is substantial for systems of medium and large sizes. By virtue of its general formulation, the LK algorithm can be used also within the class of methods that employ auxiliary basis set expansions for representing the electron repulsion integrals. © *2007 American Institute of Physics*. [DOI: 10.1063/1.2736701]

I. INTRODUCTION

The electron repulsion integral (ERI) matrix in atomic orbital (AO) basis can be decomposed by means of the Cholesky procedure.[1–4] The ERIs can then be written in the form

$$(\mu\nu|\lambda\sigma) = \sum_J^M L_{\mu\nu}^J L_{\lambda\sigma}^J, \qquad (1)$$

where M is the effective rank of the integral matrix. The Cholesky vectors are obtained by a numerical procedure whose convergence is determined by a threshold δ. Decreasing this threshold, an increasingly accurate representation of the original integrals is obtained at the price of a larger number of vectors generated. In a typical calculation, the value of M is comparable to the number of basis functions N, usually from 3 to 10 times N, much smaller than the full dimension of the integral matrix $N(N+1)/2$. Although the number of integrals that must be evaluated during a Cholesky decomposition is reduced (exactly M columns of the integral matrix are needed), the scaling is still quadratic.

The idea of decomposing two-electron integral matrix dates back to 1977 when Beebe and Linderberg[1] demonstrated the potential of the method. Recently, Koch *et al.*[4] presented an implementation suitable for large basis sets and systems, ready to be used in connection with state-of-the-art quantum chemical models.[5–10] The core of the Cholesky decomposition of the two-electron integral matrix is the exploitation of linear dependence in the product space of the atomic orbitals.[1,3] Therefore, the method can be successfully applied to molecular systems regardless of the presence of diffuse functions in the atomic basis set. The advantages of using Cholesky vectors over the explicit use of ERIs are twofold. On the one hand, a sizeable reduction in the disk space is required to store the Cholesky vectors compared to the storage requirements of the full integral matrix. On the other hand, by reformulating the basic equations of the given quantum chemical model directly in terms of Cholesky vectors, a pronounced gain in efficiency is achieved. While the latter is not a reduction of the scaling of the method, it does decrease the prefactor.

Based on the seminal work of Whitten[11] and continued by Dunlap *et al.*,[12,13] the resolution of the identity (RI) entered the scene as a convenient tool for computing four-center integrals by three-center approximations in the paper of Feyereisen *et al.*[14] Various formulations of the RI method were thoroughly discussed by Vahtras *et al.*[15] and the one based on the Coulomb metric was shown to be the most accurate. Nowadays, almost invariably, the notion of RI approximation of the ERIs refers to the use of the Coulomb metric, i.e., to the following inner projection:

$$(\mu\nu|\lambda\sigma) = \sum_{PQ} (\mu\nu|P) G_{PQ}^{-1} (Q|\lambda\sigma), \qquad (2)$$

of the r_{12}^{-1} operator onto an auxiliary basis set (labeled P and Q). The metric matrix is given by $G_{PQ}=(P|Q)$ and we have used the notation $G_{PQ}^{-1}=(G^{-1})_{PQ}$. Vahtras *et al.*[15] were also the first to point out that this formulation of the RI approximation of the ERIs is formally analogous to the Cholesky representation of Beebe and Linderberg.[1] The RI formulation as such does not prescribe the construction of the auxiliary basis set and Vahtras *et al.*[15] suggested a procedure based on manipulation of the original atomic basis set. In current implementations, however, preoptimized auxiliary basis sets[16–19] have become the widely preferred choice. The RI representation of the ERIs has become extremely popular for fitting the Coulomb energy of density functional theory

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(DFT) and for approximating ERIs for correlated calculations.[20,21] In these contexts it is also known as the density fitting (DF) approximation.[22–25] These two acronyms are used synonymously even though they both can be misleading. In fact, while it is not really a resolution of the identity that is performed, it is also important to remember that the word "density" in DF refers to orbital distributions (products of basis functions) and not to the electronic density. In this paper we will use the DF denomination.

The benefits of using the DF representation of the ERIs are identical to those mentioned in the case of the Cholesky decomposition and, not surprisingly, also the scaling of the computational cost for the evaluation of the Fock matrix for self-consistent-field (SCF) calculation is identical. If N is the number of AO basis functions and M is the number of Cholesky vectors or of auxiliary basis in DF, the cost of computing the Coulomb term formally scales as N^2M. Using prescreening in the decomposition,[4] the scaling becomes effectively quadratic with a small prefactor. When using Cholesky or DF in Hartree-Fock (HF) or hybrid DFT, however, the time-determining step is the evaluation of the exact HF exchange, a task that scales as ON^2M, where O denotes the number of occupied orbitals. For electron-rich systems and large basis sets, the quartic scaling of the evaluation of the exchange Fock matrix quickly downgrades the capacity of the standard Cholesky and DF SCF implementations[4,19] or at least renders them scarcely competitive compared to integral-direct SCF methods. The dependence of the computational scaling with the number of occupied orbitals is particularly unpleasant. To date, this *exchange problem* has been explicitly tackled only in the work of Polly *et al.*[26] for the DF approach and never discussed in the framework of the Cholesky-based SCF method. The solution proposed by Polly *et al.* is not completely satisfactory for at least two reasons. First, it is based on *ad hoc* definitions of local fitting domains with the risk of discontinuities in the potential energy surface. Second, the procedure does not yield bounded errors and, in fact, the accuracy of the result cannot be controlled solely on the basis of energy thresholds. By construction, the method of Polly *et al.* is capable of achieving asymptotic linear scaling for the evaluation of the exchange Fock matrix but the loss in terms of controllability of the accuracy of the result may appear too severe. Only very recently, the possibility of defining local fits not based on preselected domains has been suggested[24,25] but not yet investigated for reducing the scaling of the Hartree-Fock exchange in DF.

The purpose of the present work is primarily to provide a solution to the exchange problem for both Cholesky and DF SCF based on screening techniques. The spirit of this approach is therefore similar to the one pursued by Head-Gordon and co-workers in designing the integral-direct LinK scheme.[27,28] The generality of the solution we seek excludes the possibility to use a priori assumptions about the size of the system or the nature of its electronic structure. Only the short-range character of the exchange interactions will be required as working hypothesis. The result of our efforts is a scheme, named "local K" (LK), producing a sizeable reduction in the computational costs required for building any exchange-type Fock matrix from Cholesky or DF representation of the ERIs. The full control of the accuracy of the screening is the key ingredient to our solution to the exchange problem. Finally, in terms of computational demands the present implementation shows a quadratic scaling due to the intrinsic scaling of the number of Cholesky vectors.

II. THEORY

The LK algorithm is presented for the particular case of the evaluation of the exchange Fock matrix in closed shell SCF theory. A generalization to unrestricted SCF formulations is trivial, while for general exchange-type Fock matrices the algorithm would require very minute modifications, for instance, to take into account the nonpositive definite character of the exchange matrices arising from the use of specific similarity transformed Hamiltonians[5,29] and in some types of multiconfigurational SCF methods.[30] The basic ingredients to our formulation of a screened build of the exchange Fock matrix are presented in Sec. II A, while the details of the algorithm are reported in Sec. II B. Finally, the applicability of the LK scheme to the DF methods is discussed in Sec. II C.

A. The LK scheme for Cholesky representation of the ERIs

The closed shell Fock matrix in AO basis can be written in terms of the Cholesky vectors as follows:

$$F_{\lambda\sigma} = \sum_J L^J_{\lambda\sigma} \sum_{\mu\nu} P_{\mu\nu} L^J_{\mu\nu} - \frac{1}{2} \sum_{\mu J} L^J_{\lambda\mu} \sum_{\nu} P_{\mu\nu} L^J_{\nu\sigma}, \tag{3}$$

where it is clearly seen that the time-consuming part is the exchange term. In the above formulation the Coulomb term scales as N^2M, while the exchange part scales as N^3M. A straightforward way to reduce the cost of the evaluation of the exchange matrix in SCF is to rewrite the one-particle density matrix in terms of occupied molecular orbital (MO) coefficients, $P_{\mu\nu}=2\Sigma_i C_{\mu i} C_{\nu i}$, and therefore compute the exchange contribution as

$$K_{\lambda\sigma} = \sum_{iJ} L^J_{\lambda i} L^L_{\sigma i}, \tag{4}$$

where $L^J_{\lambda i}=\Sigma_\mu L^J_{\lambda\mu} C_{\mu i}$ indicates the MO half-transformed vectors. The computational effort of evaluating the exchange Fock matrix from Eq. (4) scales as ON^2M. Albeit the smaller prefactor ($O \ll N$ in most applications of interest), Eq. (4) still shows a quartic scaling with system size and poses serious limits to the range of applicability of the method. In light of this shortcoming, the design of an efficient and accurate screening of the relevant contributions to Eq. (4) is desperately needed. An attempt to provide such a tool is given in what follows.

Equation (4) can be rewritten as

$$K_{\lambda\sigma} = \sum_i (\lambda i | \sigma i). \tag{5}$$

Due to the invariance of the density under unitary transformations of the occupied orbitals, any such set of MOs can be used to compute the exchange Fock matrix. Among them, the use of localized orbitals can be advantageous once we consider Eq. (5). The integrals $(\lambda i|\sigma i)$ will show an increasing sparsity with the locality of the occupied orbitals. The possibility to screen away a substantial amount of these integrals is therefore not directly related to the size of the system; this feature is a direct consequence of the short-range nature of the exchange interaction.

In order to implement an efficient and also error bounded screening of the integrals $(\lambda i|\sigma i)$, we will make use of the following chain of inequalities:

$$|K^{(i)}_{\lambda\sigma}| = |(\lambda i|\sigma i)| \leqslant \sum_{\mu\nu} |C_{\mu(i)}||(\lambda\mu|\sigma\nu)||C_{\nu(i)}| \leqslant \sum_{\mu\nu} |C_{\mu(i)}|D^{1/2}_{\lambda\mu}D^{1/2}_{\sigma\nu}|C_{\nu(i)}| = Y^{(i)}_{\lambda}Y^{(i)}_{\sigma}, \quad (6)$$

where $D_{\mu\nu}=(\mu\nu|\mu\nu)$ are the (exact) diagonal elements of the ERI matrix in AO basis and the ith vector $Y^{(i)}$ has elements $Y^{(i)}_{\mu}=\Sigma_{\nu}D^{1/2}_{\mu\nu}|C_{\nu(i)}|$. It is important to notice that in deriving these inequalities, we have only used the fact that the ERI matrix in AO basis is positive definite and therefore satisfies the Cauchy-Schwarz inequality. The use of rigorous multipole-based integral estimates as proposed by Lambrecht and Ochsenfeld[31] could prove useful in this context but does not add substantial changes to the overall picture of our screening technique. In fact, the possibility to neglect contributions from pairs (λ,σ) in which λ and σ are sufficiently distant in space is guaranteed by the use of localized orbitals, despite the deficiency of the Cauchy-Schwarz estimates in this respect. Once again, this is a fortunate consequence of the nearsightedness of the exchange interaction.

Before describing in more detail the screening algorithm, it is important to outline two aspects. First, as shown by the inequalities in Eq. (6), the use of localized orbitals is crucial. In fact, each $Y^{(i)}$ vector will most likely retain the same sparsity as the set $C^{(i)}$. For systems sufficiently large the D matrix is also sparse, allowing an even larger degree of sparsity in the $Y^{(i)}$ compared to the MO coefficients. This is due to the fact that the diagonal ERIs are proportional to the corresponding overlap integrals, which in turn decay exponentially (in primitive basis) with the distance between the atomic centers. In other words, in the asymptotic limit, the number of significant elements in the D matrix will only increase linearly with the basis set size, with a degree of sparsity even larger than, for example, the one-particle density matrix. Second, whenever the contribution to the exchange Fock matrix from a certain number $(m<M)$ of Cholesky vectors has been computed, the inequalities can be applied in the very same way to the remainder matrix. In practice this means that the $Y^{(i)}$ vectors can be computed using updated integral diagonals, namely, $\tilde{D}_{\lambda\mu}=D_{\lambda\mu}-\Sigma^m_J(L^J_{\lambda\mu})^2$. Due to the inner projection nature of the Cholesky decomposition, these updated integral diagonals are guaranteed to remain non-negative. Most importantly, the number of significant elements in $\tilde{D}$ will decrease as more and more Cholesky vectors have already contributed to the exchange Fock matrix.

B. Details of the LK algorithm

The starting points for the LK scheme are essentially the following two facts: (1) for a given localized orbital i, only a small fraction of the entire range of σ elements contributes to a given λ; (2) the estimates $Y^{(i)}$ of Eq. (6) can be recomputed at any given time inside the loop over the Cholesky vectors by using a list of increasingly sparse (updated) integral diagonals. The update of the integral diagonals is a minor task at this stage of our discussion, as it requires at most the same amount of work needed for computing the Coulomb-Fock matrix. Setting up the $Y^{(i)}$ vectors can carry substantial overhead if this operation is performed too often. Fortunately, it turns out that an efficient screening is achieved even when the $Y^{(i)}$ vectors are updated very rarely, typically a total number of times which is about two orders of magnitude smaller than the number of Cholesky vectors M. The $Y^{(i)}$ vectors are used to get the list of σ shells contributing to a given λ shell of the $K^{(i)}$ matrix. A shell σ enters the list only if the following condition is fulfilled: $\max(Y^{(i)}_{\lambda})\cdot\max(Y^{(i)}_{\sigma})\geqslant\tau$, where τ denotes a chosen threshold and $\max(Y^{(i)}_{\lambda})$ refers to the largest value of $Y^{(i)}$ within a given shell λ. Finally, due to the fact that the exchange Fock matrix is symmetric, the range of σ can be naturally restricted as $\sigma\leqslant\lambda$.

So far, we have not taken into account the fact that prior to the evaluation of the exchange Fock matrix of Eq. (5), there is an equally costly step, namely, the MO half-transformation of the Cholesky vectors. An additional screening, specific for this task, is therefore mandatory.

As thoroughly discussed by Rubensson and Salek[32] in a similar context, identification of negligible contributions inside a matrix-matrix multiplication—balancing efficiency and the need for well-defined, strict error control—is attainable by inspection of the corresponding matrix (submultiplicative) norms. The choice of the Frobenius norm is often the most natural for matrices and has been employed in our implementation of the LK method. For a shell-pair $(\lambda\sigma)$ contribution to the matrix $K^{(i)}$, this approach leads to the following inequalities:

$$\|K^{(i)}_{\lambda\sigma}\|_F \leqslant \|L^J_{\lambda(i)}\|_F\cdot\|L^J_{\sigma(i)}\|_F, \quad (7)$$

$$\|L^J_{\lambda(i)}\|_F \leqslant \|L^J_{\lambda\sigma}\|_F\cdot\|C_{\sigma(i)}\|_F, \quad (8)$$

where the subscript F denotes the Frobenius norm,

$$\|A\|_F = \left(\sum_{ij} A^2_{ij}\right)^{1/2}. \quad (9)$$

The first of these two inequalities indicates that the (normwise) error resulting from neglect of the contribution of a shell λ to the matrix $K^{(i)}$ is proportional to the norm of the $L^J_{\lambda(i)}$ matrix. The second inequality completes the first one by providing an upper bound to the norm of the matrix $L^J_{\lambda(i)}$ and allowing a practical implementation of these inequalities within our screening algorithm. In fact, the estimate given by Eq. (8) is obtainable before performing the MO half-transformation of the Cholesky vectors.

The LK algorithm is outlined in Fig. 1. The algorithm consists in four steps. The first step is the one carrying the largest overhead and is the kernel of the entire screening. The

```
Loop over Cholesky vectors
 Loop over occupied orbitals (i)

  (1) Pre-selection and pre-ordering of the eligible shells
  Compute the $Y^{(i)}$ screening vectors from (updated) integral diagonals
  Loop over all $\mu$-shells
   If ( $MY^{(i)} \cdot max(Y_{\mu}^{(i)}) \geq \tau$ ) then
    Add the $\mu$-shell to a list $ML^{(i)}$
   EndIf
  End Loop
  Sort the $\mu$-shells in $ML^{(i)}$ by the value $max(Y_{\mu}^{(i)})$

  (2) Frobenius-norm screened MO transformation
  Loop over $\mu$-shells in the list $ML^{(i)}$
   Loop over all $\nu$-shells in the shell-pairs ($\mu\nu$) of the AO Cholesky vectors
    If ( $||L_{\mu\nu}^{J}||_F \cdot ||C_{\nu}^{(i)}||_F \geq \tau_F$ ) then
     Perform the MO half-transformation $L_{\mu(i)}^{J} = \sum_{\nu} L_{\mu\nu}^{J} C_{\nu}^{(i)}$
    EndIf
   End Loop
   Remove from $ML^{(i)}$ the $\mu$-shells that did not qualify for the MO transformation
  End Loop

  (3) Evaluation of the exchange Fock matrix $K_{\lambda\sigma}^{(i)}$
  Loop over $\lambda$-shells in $ML^{(i)}$
   Loop over the $\sigma$-shells ($\sigma \leq \lambda$) in $ML^{(i)}$
    If ( $max(Y_{\lambda}^{(i)}) \cdot max(Y_{\sigma}^{(i)}) \geq \tau$ ) then
     Compute $K_{\lambda\sigma}^{(i)} = \sum_J L_{\lambda(i)}^{J} L_{\sigma(i)}^{J}$
    Else
     Leave $\sigma$-loop
    EndIf
   End Loop
  End Loop

 End Loop
 (4) Update integral diagonals
End Loop
```

FIG. 1. Outline of the LK algorithm for reduced scaling evaluation of exchange-type Fock matrices. Notation and scaling properties are described in the text.

$Y^{(i)}$ vectors become increasingly sparse along the loop over the Cholesky vectors due to the updating of the diagonals. However, in order to balance the efficiency of the exchange Fock matrix build and increase in the screening overheads, the $Y^{(i)}$ vectors can be recomputed only a few times, indeed a number of times much smaller than M. In our implementation of the LK these two numbers differ by about two orders of magnitude in large-scale calculations without jeopardizing the overall speedup of the method. The symbol $MY^{(i)}$ in Fig. 1 represents the absolute largest element of $Y^{(i)}$. The criterium used to identify the contributing shells is based on upper bounds. Selecting the optimal value for the threshold used in this step may require some experimentation. We have found that the choice $\tau = \delta/O$, where δ is the Cholesky decomposition threshold, yields SCF energy errors on the order of δ. Thus, there is in practice no loss of accuracy compared to the standard Cholesky method. However, close to convergence of the SCF iterative procedure, it is necessary to decrease this threshold. For a smooth convergence, we have chosen to reduce τ by a factor of 100 as soon as the norm of the orbital gradient in SCF falls below δ. The second step of the LK algorithm performs a screened MO transformation of the Cholesky vectors. The efficiency of this screening depends on both the sparsity of the MO coefficients and that of the Cholesky vectors. The first is ensured by the orbital localization and the second by the fact that a given shell pair is not present in all Cholesky vectors but rather in relatively few of them. The threshold for the screening in this second step of the LK algorithm is chosen in such a way that the total error in the estimated norm of the shell-diagonal blocks of the exchange Fock matrix is bounded by $\delta:\tau_F=(\delta/OM)^{1/2}$. As for the previous step, the same damping factor is applied to this threshold in order to balance the two screenings in each SCF iteration. The third step of the LK algorithm is the contraction of the MO half-transformed Cholesky vectors to form the shell-pair blocks of the $K^{(i)}$ matrix. The preordering of the largest elements in each shell of the $Y^{(i)}$ vectors performed in step 1 allows leaving the σ loop as soon as the established condition is not met. One interesting feature of this part of the algorithm is that it intrinsically exploits the permutational symmetry of the ERIs, as can be seen by checking that only symmetry distinct contributions to the exchange Fock matrix are referenced. Finally, the fourth step is the update of the integral diagonals.

The LK algorithm does not carry any other overhead in terms of arithmetic, logical operations, or memory storage. Moreover, the build of $K^{(i)}$ in step 4 can be performed by taking full advantage of highly optimized basic linear algebra subprograms (BLAS).[33] Indeed, by a careful choice of the number of Cholesky vectors resident in memory, it is possible to reach the peak performance of the BLAS routines. Although not explicitly mentioned until now, it is clear that our screening would be penalized in case a very large number of Cholesky vectors is treated at the same time. First of all, the possibility to take advantage of the integral diagonal update more than once is reduced. Second, the screening of the MO transformation becomes less effective when the dimension of the matrices for which we compute the Frobenius norm is increased. Our tests have revealed that an optimal choice of the number of Cholesky vectors to be treated at each time is given by the number of those that share the same set of shell pairs.

The algorithm described in Fig. 1 is completely general in the sense that it can be applied to any exchange-type Fock matrix. However, for the special cases of HF and Kohn-Sham DFT, one may wish to make use of the positive definiteness of the exchange Fock matrix and exploit the Cauchy-Schwarz inequality $|K^{(i)}_{\lambda\sigma}|\leqslant K^{(i)}_{\lambda\lambda}K^{(i)}_{\sigma\sigma}$. This relation can provide a more strict criterium for the screening employed in step 1, and thus the possibility to exclude those shells μ for which $\max(Y^{(i)}_{\mu})<\tau^{1/2}$. In this way we still have a rigorous upper bound for the shell-diagonal blocks of $K^{(i)}$ but no longer for the off-diagonal blocks. However, such *bona fide* approach still gives fully controllable errors and in our experience gives further speedups with no loss of accuracy. Moreover, once the MO half-transformation has been performed (step 3), we can compute the (maximum within a shell) diagonal elements $K^{(i)}_{\lambda\lambda}$ and supply them to step 3, where the Cauchy-Schwarz inequality can be used to refine the screening criterium. These two modifications of the LK algorithm of Fig. 1 have been employed in all calculations of the present work.

C. The LK scheme for DF representation of the ERIs

In order to make the LK algorithm suitable for use in connection with the DF methods, we need to address the question of how to deal with the presence of the inverse of the metric matrix $G_{PQ}=(P|Q)$ in Eq. (2). The obvious answer is to find a way to obtain a "symmetric" DF formula of the type of Eq. (1). Several possibilities can be exploited to achieve this goal. The computationally most appealing one is by using the inverse Cholesky factor of the matrix G, namely, the (upper triangular) matrix Z that fulfills the condition $G^{-1}=ZZ^T$. This type of matrix factorization can be computed without ever referencing G^{-1} by means of a modified Gram-Schmidt orthonormalization with the inner product defined by the metric matrix G applied to a set of unit vectors spanning the column space of the matrix. The operation count and memory requirements of such a factorization for a dense matrix are similar to those of the corresponding Cholesky decomposition, and algorithms for performing this task on sparse matrices in a linear scaling fashion are available.[34,35] Once the matrix $\mathbf{Z}$ has been computed, the "DF vectors," or simply *R vectors*, $R^J_{\mu\nu}=\Sigma_P(\mu\nu|P)Z^J_P$ can be formed without the need to store the three-center ERIs and all formulas for DF can be recast in a Cholesky-type fashion.

While the construction of some kind of R vectors is not specific for the LK scheme but necessary also in the standard DF HF implementation,[19] the evaluation of the exact integral diagonals needed in step 4 of the LK algorithm in Fig. 1 represents a screening overhead. This is not the case in the Cholesky method since the integral diagonals are anyway needed during the decomposition.[1,4] However, it is by far an acceptable drawback of the LK scheme since it requires a very minor and asymptotically linear scaling computational effort. Regarding the MO half-transformation of the R vectors (step 2), the screening refinement relies on the possibility that each vector contains only a subset of all possible shell pairs. For the Cholesky vectors, this type of sparsity is the result of the decomposition procedure.[4] In the case of DF, the same could be achieved in two steps. First, by using the Cauchy-Schwarz screening for the three-center ERIs $(\mu\nu|P)$, we can derive the inequality $R^J_{\mu\nu}\leqslant D^{1/2}_{\mu\nu}V^J$ which is useful for excluding shell pairs $\mu\nu$ that are sufficiently distant ($V^J=\Sigma_P(P|P)^{1/2}|Z^J_P|$). This is analogous to the initial diagonal screening in the Cholesky decomposition of the ERIs (see Ref. 4). A similar screening of the three-center ERIs—albeit not formulated rigorously—has been already employed by Eichkorn *et al.*[16] In this step, a more fine grained screening is certainly possible by using on top of the Cauchy-Schwarz inequality the multipole-based integral estimates of Lambrecht and Ochsenfeld,[31] in order to exclude auxiliary functions centered sufficiently far away from the region spanned by a given shell pair. The second step is to identify those shell pairs that give negligible contributions to a given vector $R^J_{\mu\nu}$. Inspired by the Cholesky decomposition procedure,[4] we could exclude a given shell pair $\mu\nu$ from all subsequent ($J>m$) vectors if the ERI diagonal remainder $D_{\mu\nu}-\Sigma^m_J(R^J_{\mu\nu})^2$ falls below a chosen threshold.

III. SAMPLE APPLICATIONS

We have tested the accuracy and performance of the Cholesky LK scheme compared to both the integral-direct and the standard Cholesky[4] SCFs. We will report in a later publication the results of the application of the LK scheme to DF. As previously stated, there is practically no loss of ac-

TABLE I. Total energy (in E_h) for the Cholesky LK and the integral-direct restricted HF wave function employing correlation consistent basis sets. Values in parentheses represent the energy deviation in μE_h at.$^{-1}$.

			$(E_{LK}-E_{HF})/\delta$		
Basis set	Molecule	E_{HF}	$\delta=10^{-4}$	$\delta=10^{-5}$	$\delta=10^{-6}$
cc-pVDZ	Propyleneoxide	−191.9222094966	0.66(6.6)	1.79(1.8)	0.86(0.1)
	Benzene	−230.7219006719	1.25(10.4)	2.90(2.4)	9.00(0.8)
	Trans-cyclo-octene	−311.0748038271	2.34(10.6)	4.03(1.8)	8.25(0.4)
	Norbornenone	−344.5859142400	0.60(3.7)	2.84(1.8)	2.50(0.2)
	MDMA[a]	−629.4711318504	3.96(13.6)	6.65(2.3)	5.35(0.2)
	Stilbene	−647.2447136607	5.09(16.9)	6.68(2.2)	5.70(0.2)
	$(Alk)_1$	−40.1987065358	0.13(2.6)	0.47(0.9)	−0.18(0.0)
	$(Alk)_5$	−196.3464832756	0.69(4.0)	1.45(0.9)	1.88(0.1)
	$(Alk)_{10}$	−391.5320592167	3.00(9.4)	5.36(1.7)	4.03(0.1)
	$(Alk)_{15}$	−586.7175256192	3.83(8.1)	6.95(1.5)	5.79(0.1)
	$(Alk)_{20}$	−781.9031357160	6.00(9.6)	11.34(1.8)	7.35(0.1)
	$(Gly)_{10}$	−2144.3206781851	15.65(21.4)	29.46(4.0)	14.17(0.2)
	$(Gly)_{20}$	−4212.6290218300	32.80(22.9)	56.61(3.9)	28.21(0.2)
	$(Gly)_{30}$	−6280.9374357382	62.26(29.2)	86.40(4.0)	44.16(0.2)
	$(Gly)_{40}$	−8349.2458657351	78.82(27.8)	115.53(4.1)	
cc-pVTZ	Propyleneoxide	−191.9810559151	−0.61(−6.1)	0.67(0.7)	2.12(0.2)
	Benzene	−230.7790846292	5.47(45.6)	1.43(1.2)	2.98(0.3)
	Trans-cyclo-octene	−311.1561495541	2.58(11.7)	1.41(0.6)	4.95(0.2)
	Norbornenone	−344.6743870239	−1.70(−10.6)	1.29(0.8)	3.87(0.2)
	MDMA	−629.6425679687	−5.26(−18.1)	2.87(1.0)	7.53(0.3)
	Stilbene	−647.4067267871	−0.37(−1.2)	2.81(0.9)	8.23(0.3)
	$(Alk)_1$	−40.2131969641	−0.14(−2.8)	0.05(0.1)	0.34(0.1)
	$(Alk)_5$	−196.4020723109	−5.32(−31.2)	0.49(0.3)	2.09(0.1)
	$(Alk)_{10}$	−391.6389602259	−10.37(−32.4)	1.09(0.3)	4.18(0.1)
	$(Alk)_{15}$	−586.8757458693	−13.72(−29.2)	1.87(0.4)	5.75(0.1)
	$(Alk)_{20}$	−782.1126614933	−20.32(−32.8)	2.39(0.4)	8.04(0.1)

[a]Methylenedioxymethamphetamine.

curacy in going from the standard Cholesky SCF to the one employing the LK screening: in all our calculations the two computed energy differences with respect to the integral-direct SCF are identical within the number of digits reported in the tables. For simplicity, only the results of the Cholesky LK are explicitly shown in the tables.

All calculations have been performed with a development version of the MOLCAS quantum chemistry software.[36,37]

A. Accuracy

The accuracy of the LK screening has been investigated on a test set of molecules of varying chemical nature and size. They span a range from a few to nearly 300 atoms and include the most common elements and types of organic chemical bonds. The glycines and the alkane chains are interesting for analyzing the scaling of the error with system size, whereas others have been included for showing the applicability of the method to different local chemical environments (presence of double bonds, rings, etc.).

The most significant results are reported in Table I where the notation $(Alk)_n$ indicates the linear alkane C_nH_{2n+2} and $(Gly)_n$ a linear n-unit polypeptide glycine chain. The accuracy of the results is in most cases well within the expected Cholesky method error, i.e., within a few units of the decomposition threshold and independent of the basis set. A decrease in the accuracy of the total energy is observed for large molecules depending on decomposition threshold and basis set. The energy deviation per atom, however, is essentially size intensive with a striking invariance when the tighter decomposition thresholds are employed.

In Table II we have collected the maximum and root-mean-square (rms) deviations of the LK Cholesky from integral-direct DFT/B3LYP/SVP calculations of the activation energies for 21 of the 25 reactions of the Baker test set.[38] For the 21 selected reactions we have reoptimized the transition states and the products or the reactants at the DFT/B3LYP/6-31G* level of theory. It is clearly seen that

TABLE II. Accuracy of the LK Cholesky in the evaluation of activation energy barriers for the Baker test set of reactions. (statistics over 21 reactions). The energy deviations with respect to integral-direct DFT/B3LYP/SVP reference values are given in kcal mol^{-1}.

	Total energy		Activation energy	
δ	rms error	Max error	rms error	Max error
10^{-4}	0.1152	0.2142	0.0349	−0.0877
10^{-5}	0.0195	0.0527	0.0148	−0.0351
10^{-6}	0.0020	0.0065	0.0014	−0.0038

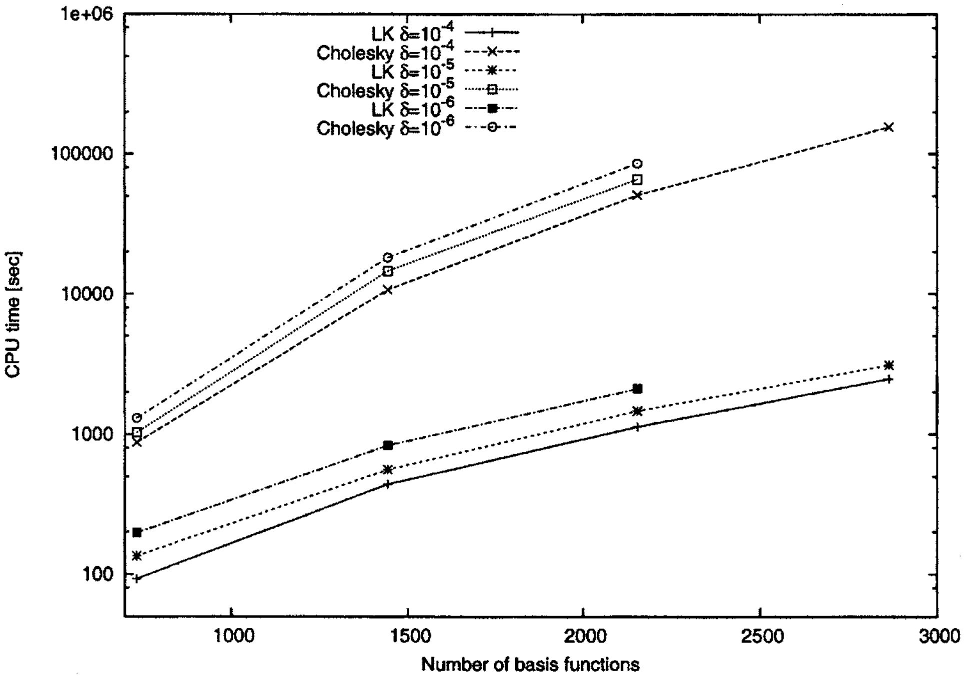

FIG. 2. Scaling behavior of the exchange Fock matrix build for linear glycines using standard and LK Cholesky with different decomposition thresholds and employing cc-pVDZ basis set. Note that a logarithmic scale has been employed.

the LK Cholesky benefits from error cancellation when computing activation energies. This is particularly evident for high thresholds (e.g., 10^{-4}), where there is a gain of one order of magnitude in accuracy compared to the one for the total energy. It should be stressed that the errors of Table II are identical to those obtainable by the standard Cholesky implementations.

B. Efficiency

In order to compare the performance of the LK algorithm with that of the standard Cholesky SCF implementation,[4] we present the timings for the construction of the exchange Fock matrix for systems of increasing size. For sake of completeness, we report also the timings for integral-direct SCF method, although comparison with the corresponding Cholesky timings is not clear-cut. In this context, in fact, integral-direct SCF refers to a nonapproximate method, while by nature the Cholesky (or DF) method is based on approximations. Using large screening thresholds in integral-direct techniques would reduce considerably the corresponding timings, perhaps producing a fairly consistent error behavior similar to the one exhibited by Cholesky and DF. However, the scope of the present article is to present an improvement on the implementations of the Cholesky and DF techniques and the comparison with integral-direct SCF is only to explore different perspectives opened by our new algorithm.

All calculations have been performed on a single AMD Opteron 2.4 GHz processor. The timings for the exchange Fock matrix construction refer to the converged HF density and the default choice of the screening thresholds for the LK, as described in Sec. II B.

In Fig. 2 the scaling of the computational time versus the number of basis functions of the standard Cholesky and LK algorithms is shown for the case of linear glycines at decreasing values of the Cholesky decomposition threshold δ. For the larger calculations the LK scheme is faster than the standard implementation by a factor from 40 to 62 and therefore a logarithmic scale is necessary to visualize both timings. The screening overheads for these calculations account in the worst case for less than 20% of the total timings reported and therefore less than 1% of the cost of the corresponding unscreened (standard) calculation. Figure 3 clarifies that the scaling of the LK algorithm is quadratic. It is instructive to notice that the screening thresholds employed in the cases investigated in Fig. 3 are not the same for all molecules. Since the default threshold τ depends on the number of occupied orbitals and τ_F also on the number of Cholesky vectors, a smaller threshold is employed for the larger chains. This is necessary in order to keep the error bounded in the total exchange Fock matrix and resulting SCF energy.

The results of Table III are an example of the onset of the exchange problem in Cholesky (and DF) SCF methods. In fact, except for the smallest glycine, the integral-direct calculation is faster than the standard Cholesky implementation as a direct consequence of the fact that the computational scaling of the exchange Fock matrix in Cholesky (and DF) methods depends on the number of occupied orbitals. The LK screening represents indeed a possible solution to the Cholesky (and DF) exchange problem, as clearly seen by comparing the LK and the integral-direct timings. It is worth noticing that in this example, we used glycine chains in the α-helix geometry to demonstrate that the LK screening is effective also for higher dimensional systems than just the linear models. One important point should be stressed at this moment. The comparison has been made with an integral-direct code which is not optimized for the evaluation of the exchange Fock matrix. Most of the standard quantum chemistry packages treat Coulomb and exchange on a similar ground, trying to reutilize the same set of two-electron integrals for performing the two tasks. However, the recent de-

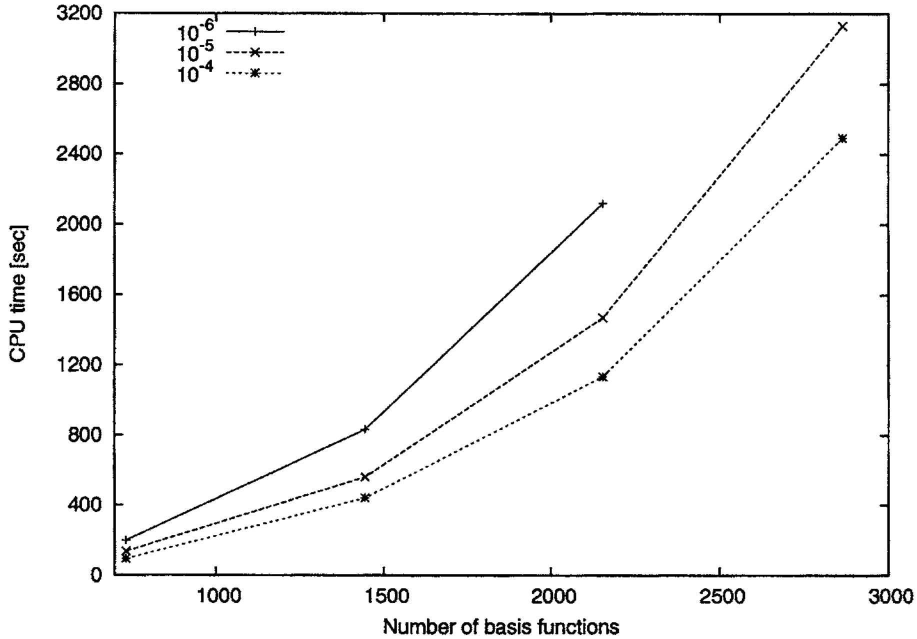

FIG. 3. Scaling behavior of the exchange Fock matrix build for linear glycines using LK Cholesky with different decomposition thresholds and employing cc-pVDZ basis set.

velopments in linear scaling techniques for computing exchange-type Fock matrices[27,28,39,40] have shown that a separate strategy specific for the exchange contribution can produce much more efficient codes, especially in the case of large insulator systems. It is fair then to admit that the use of the LK is likely to be not competitive with this new family of integral-direct approaches for extended systems and compact basis sets. In addition to that, it must be clearly stated that the LK approach is meant to improve on the present status of the Cholesky and DF based SCF methods and not to replace the well-established integral-direct techniques for a fast and linear scaling evaluation of exchange-type matrices.

Prior to the loop structure of Fig. 1, the orbital localization must be performed. At first glance this may seem an irrelevant task but actually, depending on the type of localization employed, it can become computationally dominating. This problem was also addressed by Polly *et al.*[26] who employed the Pipek-Mezey[41] localization in their algorithm for computing the exchange Fock matrix in DF. They bypassed the difficulty by performing the orbital localization only for some of the SCF iterations, thus using approximate localized orbitals in the intermediate steps. As shown in Table IV, the Pipek-Mezey localization becomes prohibitively expensive for large systems due to its scaling with the number of atoms (N_A). Other types of orbital localizations, such as the one proposed by Boys[42] and Foster and Boys[43] or the Cholesky localization recently presented by the authors,[8] appear to be suitable candidates in this respect. Indeed, the results of Fig. 4 combined with those of Table IV indicate that the latter is the choice that produces the localized orbitals (Cholesky MOs) at the lowest costs while retaining most of the efficiency of the LK screening. If not stated otherwise, all LK calculations presented in this article have been performed employing Cholesky MOs. It is also important to notice that the use of the LK screening is advantageous over the standard Cholesky algorithm even when (delocalized) canonical orbitals are employed, although at much higher cost. By comparing the highest of the curves in Fig. 4 with the lowest standard Cholesky curve in Fig. 2, we can still observe a significant speedup, ranging from a factor of about 4 in $(Gly)_{10}$ to 6 in $(Gly)_{40}$. This is a desirable feature which opens for using the LK screening in computing more general types of exchange matrices for which it may not always be possible to employ an orbital-type localization, such as for those arising in coupled cluster linear response theory[5] and in certain applications of the multiconfigurational SCF methods.[30]

TABLE III. CPU time (in seconds, AMD Opteron 2.4 GHz processor) for the exchange Fock matrix build using the cc-pVDZ basis set. The decomposition threshold is $\delta = 10^{-4}$.

Molecule[a]	O	N	M	Method: Integral direct	Cholesky	LK Cholesky
$(Gly)_{10}$	155	734	3 556	1 395	933	216
$(Gly)_{20}$	305	1444	6 957	7 059	12 680	1038
$(Gly)_{30}$	455	2154	10 348	17 019	56 305	2705

[a] α-helix geometry used for the glycines.

TABLE IV. CPU time (in seconds, AMD Opteron 2.4 GHz processor) for the various localization methods based on the restricted HF/cc-pVDZ optimized wave function. For Cholesky localization the time includes the calculation of the density matrix from canonical orbitals. N_A is the number of atoms.

Molecule[a]	O	N	N_A	Localization: Pipek-Mezey[b]	Boys[b]	Cholesky
$(Gly)_{10}$	155	734	73	78	2	0.1
$(Gly)_{20}$	305	1444	143	1432	20	1.0
$(Gly)_{30}$	455	2154	213	8895	105	3.5
$(Gly)_{40}$	605	2864	283		270	7.7

[a] Linear geometry used for the glycines.
[b] Cholesky MOs used as initial orbitals.

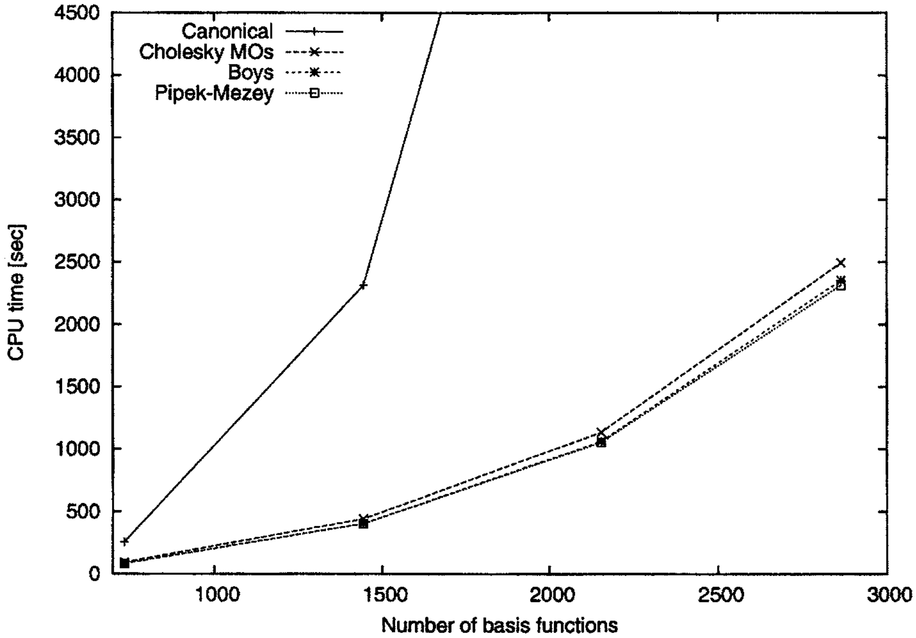

FIG. 4. Scaling behavior of the exchange Fock matrix build for linear glycines using LK Cholesky ($\delta = 10^{-4}$) with different types of localized orbitals and employing cc-pVDZ basis set. The number of Cholesky vectors for the four glycines are, respectively, $M_{10}=3552$, $M_{20}=6974$, $M_{30}=10\,344$, and $M_{40}=13\,774$.

Finally, for the linear alkane chains the scaling of the LK method compared to those of the standard Cholesky and integral-direct SCFs are shown in Fig. 5. Here the onset of the exchange problem is delayed by the relatively small number of occupied orbitals compared to the basis set size (cc-pVTZ basis). The LK algorithm substantially reduces the computational cost for the Cholesky (and DF) SCF evaluation of the exchange Fock matrix, thus reducing the gap to linear scaling integral-direct algorithms[27,28] for systems containing up to perhaps a few hundred atoms and noncompact basis sets.

C. Dinuclear diazene iron complex

We will briefly present an application of our LK screening algorithm to a molecular system to be considered as a prototype of those for which the advantages of using the LK are substantial over both standard Cholesky and integral-direct SCF methods. The molecule in question is a dinuclear diazene iron complex, a compound of interest because it represents a suitable model for studying nitrogenase activity (see Ref. 44 and references therein for details). For the scope of the present article, however, the interesting properties of this system are (1) it is electron rich, (2) it contains many different types of atoms, and (3) that the transition metals are not directly connected to each other but sit in two different moieties of the complex (see Fig. 6). The first property ensures early onset of the exchange problem, the second is a challenge for the accuracy of the LK, and the third should allow the LK to perform an efficient screening which could ultimately decouple the exchange contributions of the two moieties. If this is the case, the LK would produce significant speedups in the calculation of the exchange Fock matrix,

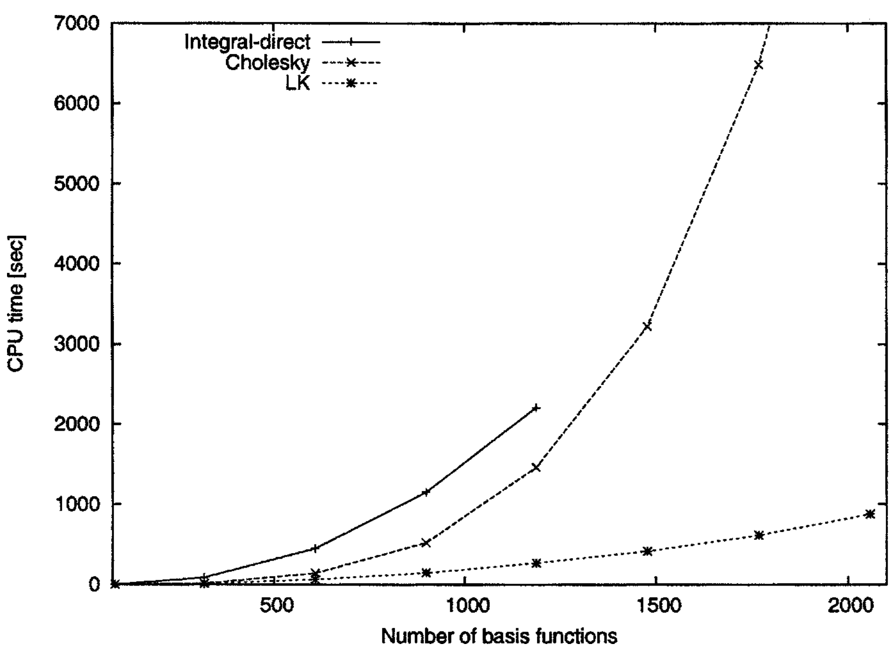

FIG. 5. Comparison of the scaling behavior of the exchange Fock matrix build for linear alkanes using standard Cholesky, LK Cholesky, and integral-direct methods. Decomposition threshold $\delta=10^{-4}$ and cc-pVTZ basis set.

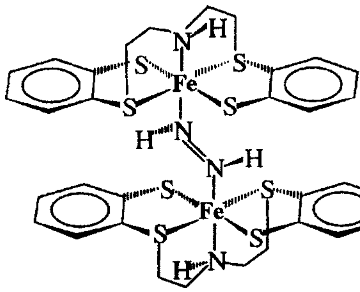

FIG. 6. Schematic structure of the dinuclear diazene iron complex used as sample application for the LK scheme.

TABLE VI. Total energy (E_h) for the Cholesky LK and the intgral-direct restricted HF wave function of the dinuclear diazene iron complex. The deviations between the two methods are reported in units of the ERI Cholesky decomposition threshold $\delta=10^{-4}$. Values in parentheses represent the energy deviation in μE_h at.$^{-1}$.

Basis set[a]	E_{HF}	$E_{LK}-E_{HF}$	
3-21G	−7131.684 493 576 5	4.1	(5.0)
cc-pVDZ	−7155.505 109 588 1	40	(48.8)
cc-pVTZ	−7155.986 904 159 8	31	(37.8)

[a]The basis set on the iron atoms is ANO-L ($5s4p2d1f$).

demonstrating the applicability of Cholesky and DF based hybrid DFT and post-HF methods to the field of bioinorganic chemistry.

We have performed restricted HF calculation on the diazene iron complex of Fig. 6 using the structure provided by Reiher *et al.* (complex 1 of Ref. 44). The basis set on the iron atoms is the ANO-L set contracted as $5s4p2d1f$ and has been kept fixed for all calculations. The quality of the basis set on each of the remaining atoms has been varied (the 3-21G, cc-pVDZ, and cc-pVTZ basis sets were used). As shown in Table V, the scaling of the computational cost of the LK method is much less severe than that of the integral-direct method and, in particular, of the standard Cholesky method. The largest speedup of the exchange Fock matrix build is approximately a factor 4 compared to the former and 5 compared to the latter. Here it is important to note that the speedup is achieved for a system that is relatively compact. In a sense, we have also supplied evidence for an earlier statement given in this article regarding the fact that the potential of the LK algorithm is not directly related to the size of the system.

Finally, the results of Table VI show that the LK screening not only makes the evaluation of the HF exchange significantly faster but also achieves this goal while keeping the expected accuracy of the Cholesky method.

IV. A POSSIBLE ROUTE TO LINEAR SCALING LK

As shown by the calculations on the test set chosen in the present article, the LK scheme reduces the computational scaling of the exchange Fock matrix build in Cholesky SCF from quartic to quadratic. While this is an important achievement that significantly expands the application range of the Cholesky and DF methodologies, it does not make them competitive with integral-direct linear scaling techniques[27,28] whenever the system size goes beyond the 100 atom regime and relatively compact atomic basis sets are used. An improved linear scaling LK scheme is therefore highly desirable, although a fully linear scaling Cholesky SCF will still not be possible with the present implementations of the ERI Cholesky decomposition.[4] Moreover, when discussing linear scaling algorithms one should in principle make sure that every step can be performed with a number of arithmetic and logical operations, and preferably memory requirements, scaling linearly with system size. In the present discussion, however, we will focus only on the exchange Fock matrix build (step 3) and the MO half-transformation of the Cholesky vectors (step 2) in the LK, which are the most computationally demanding steps of the entire algorithm.

TABLE V. CPU time (in seconds, AMD Opteron 2.4 GHz processor) for the exchange Fock matrix build of the dinuclear diazene iron complex (82 atoms, 218 occupied orbitals).

			Method		
Basis set[a]	N	M	Integral direct	Cholesky	LK Cholesky
3-21G	568	2539	613	382	158
cc-pVDZ	896	4202	2244	2081	555
cc-pVTZ	1924	9503	15827	19958	3698

[a]The basis set on the iron atoms is ANO-L ($5s4p2d1f$).

The LK algorithm is quadratic by construction and not because of a poor implementation. While Eq. (5) represents a linear scaling formulation of the exchange as a sum of localized orbital contributions, Eq. (4) gives rise to quadratic scaling even in a localized orbital basis, as the total number of Cholesky vectors scales linearly with system size (see Ref. 4). It is therefore clear that efforts for reducing the scaling of the LK must be directed toward a reduction of the range of the J index spanned by each fixed orbital distribution $\mu\nu$ in the Cholesky vectors or their DF counterparts. The long-range decay of the Coulomb interaction is the origin of the relatively dense nature of the Cholesky vectors. The same also holds for the DF R vectors due to the particular choice of metric (see, e.g., Ref. 24 for details). Exploiting that Cholesky decomposition and DF are both inner projections of the ERI matrix,[1,15] the most promising route toward linear scaling LK is to combine it with the technique presented by Sodt *et al.*[25] for a DF formulated in terms of local fitting domains. The locality imposed to the metric matrix G by this approach effectively scatters the otherwise dense linear algebra involved in the construction and use of the R vectors. The approach of Sodt *et al.* has not yet been confronted with the problem of fitting the exchange energy, for which it appears as a natural generalization of the one proposed by Polly *et al.*,[26] with also the advantage of a guaranteed continuity of the resulting potential energy surface. However, the accuracy of this approach is still entrusted solely to a heuristic definition of the extent of the fitting domains, reducing the possibility for a smooth control of the errors. For its optimal use within the LK scheme, it is therefore desirable to complement it with more robust localization criteria based directly on the closeness between the local and the global fitting.

We will return to the subject of a linear scaling formulation of the LK algorithm in forthcoming publications.

V. CONCLUDING REMARKS

We have introduced a new algorithm, local K or simply LK, for the evaluation of the exchange Fock matrix when using Cholesky decomposed ERIs. The LK screening algorithm makes use of rigorous estimates of the orbital contributions to the exchange Fock matrix and avoids *ad hoc* partitioning of the molecule in local domains. It therefore allows for strict error control. Localization of the occupied orbital space is a crucial ingredient, although the LK is faster than the standard Cholesky implementation even if canonical orbitals are employed. This feature is of interest for the application of the method to the computation of more general exchange-type matrices arising, for example, in multiconfigurational SCF theory.

Employing the LK scheme, the computational scaling of the exchange Fock matrix build in Cholesky SCF is shown to be reduced from quartic to quadratic with negligible screening overheads (compared to the standard implementation) and virtually no loss of accuracy. As the construction of the exchange Fock matrix is the bottleneck of a Cholesky SCF iteration, the use of the LK screening allows for speedups of about one to two orders of magnitude compared to the standard Cholesky SCF implementation. The LK scheme can be used directly in connection with the DF approximation.

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